

🔗 Guided Lab 2.1 — Introduction to Supervised Learning with Scikit-learn

🎯 Lab Goal

In this guided lab, students will build and test a basic supervised machine learning model using Python and Scikit-learn. Students will explore the workflow of loading data, training a model, generating predictions, and evaluating performance.

📖 Learning Objectives

By completing this lab, students will be able to:

- Load and explore a dataset in Python
- Split data into training and testing datasets
- Train a supervised learning model using Scikit-learn
- Generate predictions from a trained model
- Evaluate model performance using accuracy metrics

💻 Tools and Technologies

- Python
- Jupyter Notebook / Google Colab
- Pandas
- Scikit-learn
- NumPy

📁 Dataset Information

Dataset Name:

`cybersecurity_login_dataset.csv`

Features:

- `login_attempts` → Number of login attempts
- `failed_logins` → Number of failed login attempts

- `unusual_login_hour` → Whether login occurred at an unusual hour (0 = No, 1 = Yes)
- `new_ip_address` → Whether login came from a new IP address (0 = No, 1 = Yes)

Target Variable:

- `suspicious`
 - 0 = Normal Activity
 - 1 = Suspicious Activity

✂ Step 1 — Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
```

✂ Step 2 — Load the Dataset

```
# Load dataset
df = pd.read_csv("cybersecurity_login_dataset.csv")

# Display first 5 rows
df.head()
```

✂ Step 3 — Explore the Dataset

```
# Dataset information
print(df.info())

# Check for missing values
print(df.isnull().sum())

# Display dataset shape
print(df.shape)
```

✂ Step 4 — Define Features and Target

```
# Features
X = df.drop("suspicious", axis=1)

# Target variable
y = df["suspicious"]

print(X.head())
print(y.head())
```

✂ Step 5 — Split the Data

```
# Split dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print(X_train.shape)
print(X_test.shape)
```

✂ Step 6 — Train the Logistic Regression Model

```
# Create model
model = LogisticRegression()

# Train model
model.fit(X_train, y_train)
```

✂ Step 7 — Generate Predictions

```
# Predict on testing data
y_pred = model.predict(X_test)

# Display first predictions
print(y_pred[:10])
```

✂ Step 8 — Evaluate the Model

```
# Accuracy score
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# Confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:\n", cm)

# Classification report
print(classification_report(y_test, y_pred))
```

✂ Step 9 — Interpret the Results

Questions to Answer

1. What was the model accuracy?
2. What do the confusion matrix results tell you?
3. Which features may help identify suspicious behavior?
4. How can machine learning support cybersecurity monitoring?



Example Output

Accuracy: 0.84

Confusion Matrix:
[[38 12]
 [7 63]]



Reflection

Write a short paragraph explaining:

- What the model predicted well
- What challenges or limitations you observed
- Why supervised learning is useful in cybersecurity



Submission Requirements

Submit:

- Completed Jupyter Notebook (.ipynb)
- Screenshots of your output (if requested)
- Short reflection paragraph



Deliverable

Completed Guided Lab 2.1 Notebook Submission